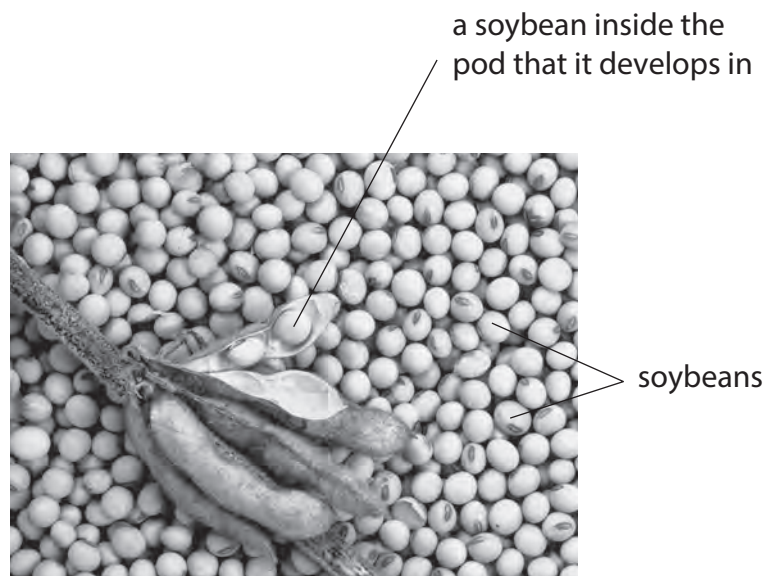


6 The photograph shows some soybeans.



© inga spence / Alamy Stock Photo

- (a) Eating soybeans may reduce cholesterol and low-density lipoproteins (LDLs) in the blood.

Explain why eating soybeans may reduce the risk of heart disease.

(3)

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- (b) The extent to which soybeans reduce cholesterol depends on the variety of soybeans. This reduction is influenced by two proteins found in soybeans, glycinin and β -conglycinin.

Give **two** ways that the primary structure of two proteins could differ from each other.

(1)

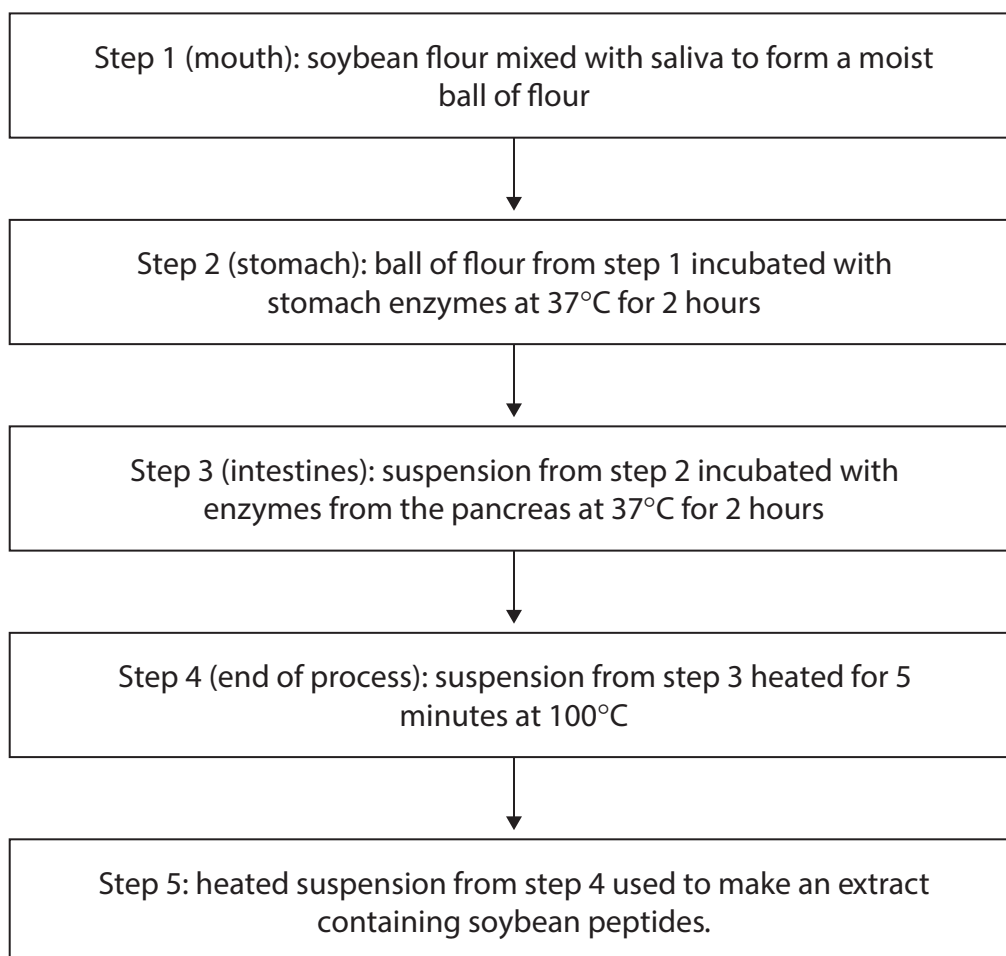
1

2

- (c) The effect of soybean extracts on cholesterol production by liver cells growing in culture was investigated.

Scientists produced fat-free soybean flour and digested it with enzymes, in a way that copied digestion in the human digestive system.

The diagram shows some of the steps used.



- (i) Explain why the scientists used digested soybeans to make the extract for this investigation.

(2)

- (ii) Suggest why the soybean flour was incubated with different enzymes in steps 2 and 3.

(1)

- (iii) Explain why the soybean flour and enzymes were incubated at 37 °C for 2 hours in steps 2 and 3.

(2)

37 °C

2 hours

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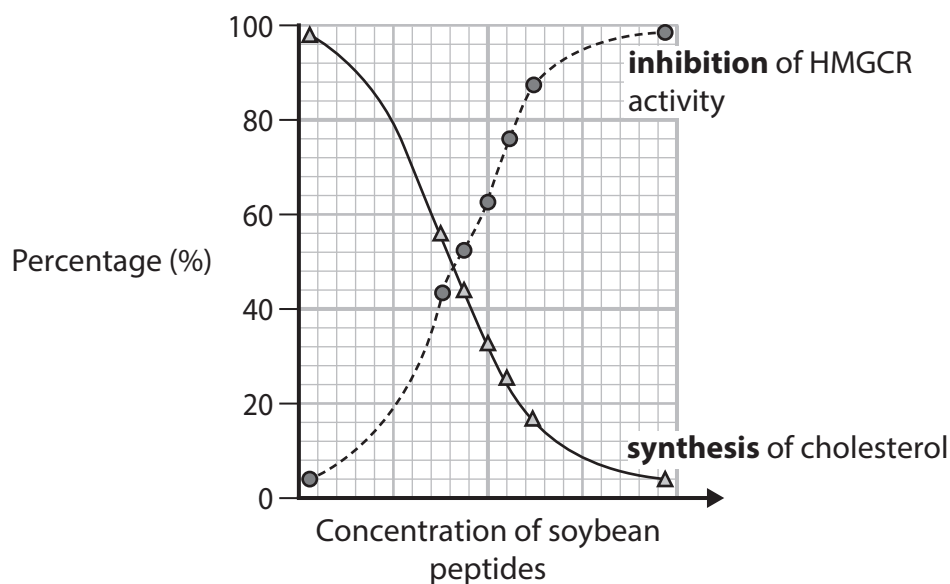


- (d) The soybean peptides, at different concentrations, were incubated with the liver cells growing in culture.

The enzyme HMGCR is involved in the synthesis of cholesterol.

The effect of soybean peptides on the activity of this enzyme and the synthesis of cholesterol was measured.

The graph shows the results.



Explain why eating soybeans may reduce cholesterol levels in the blood.

Use the information in the graph to support your answer.

(3)

(Total for Question 6 = 12 marks)

